

Ethics without IRB, is that possible? The Case Study of Participatory Sessions with ASD Children in Mexico

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ABSTRACT

Research involving participatory design with children necessarily implies a careful review of the research methods to ensure participants' safety; However, in Latin America may be different than in the Global North countries. Over the past 13 years, we have partnered with a school-clinic in Mexico that specializes in assisting children with autism. During this time, we designed, developed, and assessed UbiComp technology to assist in various stages of a child's therapy cycle. We have been collaborating with the school-stakeholders such as children and young adults with autism, parents, teachers, psychologists, and therapists, including them in all the phases of the projects. In this paper, we reflect on our experiences at this school-clinic, discussing the formality and ethics when conducting research. We close by presenting the implications we have learned while conducting research with this population.

CCS CONCEPTS

• **Human-centered computing** → Ubiquitous and mobile computing; Empirical studies in ubiquitous and mobile computing.

KEYWORDS

LATAM studies, Participatory design, children with autism

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1 INTRODUCTION

Research involving children must follow ethical guidelines to guarantee their safety, privacy, and security. Therefore, in most Universities (mostly from the Global North), ethical committees' boards review and certify that the methodologies followed by researchers

fulfill those requirements. For example, in the United States and Canada, the Institutional Review Board (IRB) from each university provides guidelines and standards to ensure research follows regulations formulated and enforced by various government departments, including the Department of Health and Human Services. Moreover, researchers who work with human subjects must periodically conduct training and courses usually provided by the Collaborative Institutional Training Initiative (CITI). As a result of this process, researchers get approved for the research protocol, compensation, flyers, informed consent (from caregivers when working with minors), and assent forms (for children).

However, none all countries/universities follow the same "formality" and "standardization" as those processes need financial resources (e.g., to pay for the training and the time of the review board members) and well-established guidelines from government and Departments of Health, putting in disadvantage to countries in terms of research and collaboration. This is especially important when working with vulnerable populations like children. Researchers usually "mimic" and adapt formal practices of the IRB to maintain the integrity and ethical principles of the research.

In this paper, we present a set of ethical practices and strategies followed by the authors over the past 13 years at Pasitos, a school clinic specializing in assisting children with autism, by conducting participatory design and deployment of digital health interventions for ASD (Autism Spectrum Disorder) children. This workshop represents an opportunity to first, present those practices to a broader audience from both, the Global North and the Global South, discuss how they could be improved and applied in similar contexts, and propose how those practices should be adapted with the use of emerging technologies for neurodiverse children.

2 THE CASE OF PASITOS

Pasitos is a school-clinic in Tijuana, México specializing in assisting ASD children with autism. Pasitos provides education programs for ASD children between the ages of 3 and 20 years old, besides other therapy services. Most ASD children have severe symptoms across the spectrum, and they get training for behavior management and learn about self-care, language development, and communication and social skills.

Over the past 13 years, we have been developing technology such as augmented reality to support social interaction and cognition, ambient displays to promote positive behaviors, tangible computing for fun and interactive therapies, brain-computer interfaces, and games to support neurofeedback treatments and executive function training, that have been deployed at the school-clinic (see examples in Table 1).

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Table 1: UbiComp technology designed and developed with/for ASD children at Pasitos.

Project	UbiComp technology	Aim
MOBIS [1.]	Augmented reality	Object discrimination/identification training
MOSOCO [2.]	Augmented reality	Improve social skills in real-life situations
FarmerKeeper [3.], [4.]	Brain-computer interfaces and videogames	Neurofeedback training helping to improve attention
Sound preferences [5.]	Wearable brain-computer headband	Understand the appropriate sounds to be used in music-based interventions for children with autism
Hunting Relics [6.]	Exergames/ Interactive floor	Practicing of motor coordination exercises
Bx-Balloons [7.]	Ambient display	Behavior awareness
BendableSound [8.][9.][10.]	Elastic display	Support music therapy session
Circus in Motion [11.]	Exergame	Vestibular system
Ethereal Breathing [12.]	Holographic display	Breathing Exercises
Sonification of movements [13.]	Wearables	Upper limb movements


Figure 1: Setting up a research project with the Pasitos' staff.

3 ETHICS GUIDELINES

In this section, we reflect on our experiences with the ethical guidelines we follow when conducting research in this context.

3.1 Getting the approval.

We first approached the administrative members of Pasitos, asking them to allow us to learn from them to create technology with, for and by them. We shared our goals and together defined a mutual agreement of benefit. Then for each project, together, we developed guidelines.

3.2 Ethical guidelines and procedures followed while conducting participatory design.

The research team created the protocol for the study that included study design, protocols of ethics, and assent of participants. Then, we shared the protocol with the Pasitos team (Figure 1). Mimicking the IRB, a set of Pasitos staff, including the administrative staff, the school principal, and the coordinators, that will not participate

in the study review, requested changes to the protocols provided, and reviewed the wording of the consent formats to make sure it is appropriate for participants. Together, the research team and Pasitos staff set the calendar for the study, where the first task is setting up a meeting (informal group meeting) with teachers, therapists, and administrative staff to invite them to participate in the research.

After this initial meeting, when the research activities are observation, interviews, and/or workshops with the teacher, attending those meetings voluntarily means consent.

Then, by the suggestion of Pasitos staff, we created a simplified version of the consent forms so parents could easily understand the research, and if they would like their children to participate, they sign and select if we could also record audio and videos. We used to use parents as proxies [14.], given the ASD symptoms across the spectrum tend to be severe for most children. On average, the 3-pages consent forms for parents include a description of the project, the UbiComp technology that will be evaluated, the study design and description of their children's participation, the data collection and how these will protect the children's privacy. Despite our formality, most parents agreed to participate in the project if the research procedure was already approved by the Pasitos staff, showing their trust in the institution. Only a few parents per project request more details. In most of the projects we deployed, all parents signed the consent form, either in an informal meeting or when picking up or dropping off their children at Pasitos.

Then we start the collection of data that could be from interviews, participatory workshops, and direct observations. We audio-record interviews and participatory workshops, and direct observations are video-recorded, or we take notes.

3.3 Ethical guidelines and procedures followed while conducting evaluations.

During the evaluation stage of the study, we set up another informal meeting with the Pasitos staff to share and discuss the evaluation procedure. During this meeting, we presented in depth how children will interact with the UbiComp technology, so we can agree and ensure their safety.

Every time children participate, teachers acted as “proxies” to assent as they know better when they are willing or not to conduct an activity.

3.4 Ethical guidelines and procedures presenting results and follow-ups

After the study finished and the data was analyzed, and even before publishing the results, we conducted a meeting with the teachers, staff, and board to present the findings. If the findings are positive and the UbiComp technology has the potential to continue using it, we train teachers and staff to use it as long as possible. Unfortunately, technology transfer has been difficult as the economic resources and professional experts are limited on both sides, research and Pasitos.

When research is published, we carefully review that data is anonymized, pictures presented do not show children’s faces, and we used the wording and vocabulary used and accepted by participants; for example, we have been mainly treating the disability as a third person (i.e., individuals with autism).

4 DISCUSSION

Given our experiences conducting research studies with vulnerable populations in the Mexican context. In the following sections, we present the ethical implications we have learned while conducting research.

4.1 Trade-off between providing context and protecting participant privacy

When conducting research studies, enough information should be provided to understand the context but without revealing participants’ information that can help identify them. In studies in the Majority World (i.e., Global South), researchers often face challenges as publications usually request enough details and even positional statements and details that could potentially identify participants, especially when sample sizes are smaller and working with minorities and individuals with disabilities. Therefore, balancing, providing necessary information, and protecting participant identities are crucial.

4.2 Contextualizing guidelines for disabilities and individuals facing barriers.

The field of Human-Computer Interaction, UbiComp, Accessibility, and technology is constantly evolving. The more we understand our context, the more diverse and inclusive we are over time. Guidelines are constantly provided to properly discuss topics in this field, mainly on the Global North. However, it is essential to contextualize those guidelines given the cultural and background of participants, as they may differ from the Global North guidelines. For example, using first person or third-person language when referring to a disability may vary across contexts. More privileged areas tend to use first-person language when referring to Autism (e.g., Autistic person), and they provide a well-founded reason behind this choice. On the other hand, in different contexts, they may find it more appropriate to use a third-person language (e.g., person with Autism). Respecting participants’ preferences and adapting to their

beliefs based on their context and reality, which may be influenced by resource availability, is crucial, particularly when dealing with individuals with less privilege.

4.3 Challenges in Obtaining Informed Consent

Obtaining informed consent from participants who may not be familiar with research can be challenging. Participants from privileged areas are often more aware of what a Ph.D. is and understand the implication and process of research. However, in less privileged contexts, a lack of guidance and appropriate information may hinder trust-building. To address this, researchers must invest extra effort in explaining the meaning of research, the technology to be used, and the potential outcomes. Participants then may have either high expectations and/or misconceptions about the potential impact of the technology. Therefore, a comprehensive explanation of the research process, alongside simplifying the consent form, can help foster trust among participants and encourage them to participate for the sake of advancing scientific knowledge rather than personal gain.

5 CHALLENGES OF SUSTAINING TECHNOLOGY BEYOND THE STUDY

In studies involving technological solutions, participants may find value in the technology and express a desire to continue using it beyond the research project. However, the reality is that prototypes often have a limited lifespan and need technical and financial support to keep them updated and working properly. Therefore, despite efforts to leave the technology in place, challenges arise in the long term, and participants eventually stop using it. Addressing this challenge requires exploring sustainable approaches to technology development, considering the context and available resources to ensure that the benefits of technology reach participants beyond the short term.

6 CONCLUSION

In conclusion, conducting studies in the Global South and engaging with participants facing barriers or disabilities present unique challenges. Balancing the need for context and privacy, contextualizing guidelines, building trust through informed consent, and addressing the sustainability of technological solutions are crucial steps toward overcoming these challenges and conducting ethical research in diverse contexts.

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REFERENCES

- [1.] Franceli L. Cibrian, Judith Ley-Flores, Joseph W. Newbold, Aneesha Singh, Nadia Bianchi-Berthouze, and Monica Tentori. 2021. Interactive sonification to assist children with autism during motor therapeutic interventions. *Personal and Ubiquitous Computing* 25, 2: 391–410. <https://doi.org/10.1007/S00779-020-01479-Z/METRICS>
- [2.] Franceli L. Cibrian, Melisa Madrigal, Marina Avelais, and Monica Tentori. 2020. Supporting coordination of children with ASD using neurological music therapy:

- A pilot randomized control trial comparing an elastic touch-display with tambores. *Research in Developmental Disabilities* 106: 103741. <https://doi.org/10.1016/J.RIDD.2020.103741>
- [3.] Franceli L. Cibrian, Jose Mercado, Lizbeth Escobedo, and Monica Tentori. 2018. A step towards identifying the sound preferences of children with autism. *ACM International Conference Proceeding Series* 158–167. <https://doi.org/10.1145/3240925.3240958>
- [4.] Franceli L. Cibrian, Oscar Peña, Deysi Ortega, and Monica Tentori. 2017. BendableSound: An elastic multisensory surface using touch-based interactions to assist children with severe autism during music therapy. *International Journal of Human-Computer Studies* 107: 22–37. <https://doi.org/10.1016/J.IJHCS.2017.05.003>
- [5.] Franceli L. Cibrian, Monica Tentori, and Ana I. Martínez-García. 2016. Hunting Relics: A Persuasive Exergame to Promote Collective Exercise in Young Children. <https://doi.org/10.1080/10447318.2016.1136180> 32, 3: 277–294. <https://doi.org/10.1080/10447318.2016.1136180>
- [6.] Lizbeth Escobedo, David H. Nguyen, Lou Anne Boyd, Sen H. Hirano, Alejandro Rangel, Daniel García-Rosas, Monica Tentori, and Gillian R. Hayes. 2012. MOSOCO: A mobile assistive tool to support children with autism practicing social skills in real-life situations. *Conference on Human Factors in Computing Systems - Proceedings*: 2589–2598. <https://doi.org/10.1145/2207676.2208649>
- [7.] Lizbeth Escobedo, Mónica Tentori, Eduardo Quintana, Jesus Favela, and Daniel Garcia-Rosas. 2014. Using augmented reality to help children with autism stay focused. *IEEE Pervasive Computing* 13, 1: 38–46. <https://doi.org/10.1109/MPRV.2014.19>
- [8.] Foad Hamidi, Melanie Baljko, and Isabel Gómez. 2017. Using participatory design with proxies with children with limited communication. *ASSETS 2017 - Proceedings of the 19th International ACM SIGACCESS Conference on Computers and Accessibility*: 250–259. <https://doi.org/10.1145/3132525.3132527>
- [9.] Jose Mercado, Lizbeth Escobedo, and Monica Tentori. 2020. A BCI video game using neurofeedback improves the attention of children with autism. *Journal on Multimodal User Interfaces 2020 15:3* 15, 3: 273–281. <https://doi.org/10.1007/S12193-020-00339-7>
- [10.] Jose Mercado, Ismael Espinosa-Curiel, Lizbeth Escobedo, and Monica Tentori. 2019. Developing and evaluating a BCI video game for neurofeedback training: the case of autism. *Multimedia Tools and Applications* 78, 10: 13675–13712. <https://doi.org/10.1007/S11042-018-6916-2/METRCS>
- [11.] Ivonne Monarca, Monica Tentori, and Franceli L. Cibrian. 2023. Understanding the musical interaction of children with autism spectrum disorder using elastic display. *Personal and Ubiquitous Computing*: 1–18. <https://doi.org/10.1007/S00779-022-01703-Y/METRCS>
- [12.] Arturo Morales, Franceli L. Cibrian, Luis A. Castro, and Monica Tentori. 2022. An adaptive model to support biofeedback in Aml environments: a case study in breathing training for autism. *Personal and Ubiquitous Computing* 26, 6: 1445–1460. <https://doi.org/10.1007/S00779-020-01512-1/METRCS>
- [13.] Oscar Peña, Franceli L. Cibrian, and Monica Tentori. 2021. Circus in Motion: a multimodal exergame supporting vestibular therapy for children with autism. *Journal on Multimodal User Interfaces* 15, 3: 283–299. <https://doi.org/10.1007/S12193-020-00345-9/METRCS>
- [14.] Monica Tentori, Lizbeth Escobedo, Carlos Hernandez, Aleksandar Matic, and Gillian R. Hayes. 2016. Pervasive Displays in Classrooms of Children with Severe Autism. *IEEE Pervasive Computing* 15, 3: 48–57. <https://doi.org/10.1109/MPRV.2016.49>